Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “friction” and “static” to compare your model or result. Sentence starters are optional.

friction: a contact force opposing relative sliding or its tendency

static: not sliding relative to the contacting surface

kinetic: associated with motion; kinetic friction acts during sliding

01 Bring back an earlier idea

A runner pushes the ground backward with 50 N.
Name the partner force, its size and the object it acts on.
Explain its role in the runner’s motion.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Below the limit

The box stays at rest with an 8 N push right. Maximum static friction is 15 N.
Find actual friction and net force.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. At rest; static limit 15 N

Marzano check - circle: ① ② ③

04 Crossing the limit

For the same box, the push increases to 18 N. Sliding friction is 12 N. Once it slides right, find net force.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Static and kinetic friction

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Can friction help?

Your foot pushes backward on the ground without slipping. Which way does ground friction act on your foot? **Explain** why“friction always points backward”is wrong.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

A cart starts at 4 m and moves right at 6 m/s for 3 s. **Predict** final position. Allowed endpoints are 20 to 24 m; state this interval and one assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

07 Exit check

Maximum static friction is 20 N.A 12 N push does not move the box.
Find actual friction. A 25 N push starts sliding with 16 N kinetic friction: find net force.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A box has maximum static friction 8 N and kinetic friction 6 N.
Predict the behavior for pushes of 4, 8 and 10 N.
Draw the sliding case.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3